

Stat 654 - Homework 5

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Problem 1: Gaussian couplings

Recall that a coupling of two random variables with given marginals means a joint construction of them on the same probability space. Let $X, Y \sim N(0, 1)$ be two independent $N(0, 1)$ random variables.

1. Let $Z := aX + bY$. You may use without proof that Z is also Gaussian. Compute the mean and variance of Z . For what condition on a and b does Z have distribution $N(0, 1)$?

Solution: Let $Z := aX + bY$ with $X, Y \sim N(0, 1)$ independent. Then

$$\mathbb{E}[Z] = a\mathbb{E}[X] + b\mathbb{E}[Y] = 0.$$

Also,

$$\text{Var}(Z) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

Since X and Y are independent, $\text{Cov}(X, Y) = 0$, and since $\text{Var}(X) = \text{Var}(Y) = 1$, we get

$$\text{Var}(Z) = a^2 + b^2.$$

Therefore $Z \sim N(0, 1)$ if and only if

$$a^2 + b^2 = 1.$$

2. Show that $(X, \rho X + \sqrt{1 - \rho^2} Y)$ is a coupling of two standard Gaussian random variables for any $\rho \in [-1, 1]$. Draw a scatterplot of the pair and plot histograms of both coordinates to numerically verify that each marginal distribution is standard Gaussian. This coupling is called the identical coupling when $\rho = 1$, the independent coupling when $\rho = 0$, and the antithetic coupling when $\rho = -1$.

Solution: Define

$$Z_\rho := \rho X + \sqrt{1 - \rho^2} Y, \quad \rho \in [-1, 1].$$

Then

$$\mathbb{E}[Z_\rho] = \rho \mathbb{E}[X] + \sqrt{1 - \rho^2} \mathbb{E}[Y] = 0,$$

and

$$\text{Var}(Z_\rho) = \rho^2 \text{Var}(X) + (1 - \rho^2) \text{Var}(Y) + 2\rho\sqrt{1 - \rho^2} \text{Cov}(X, Y) = \rho^2 + (1 - \rho^2) = 1,$$

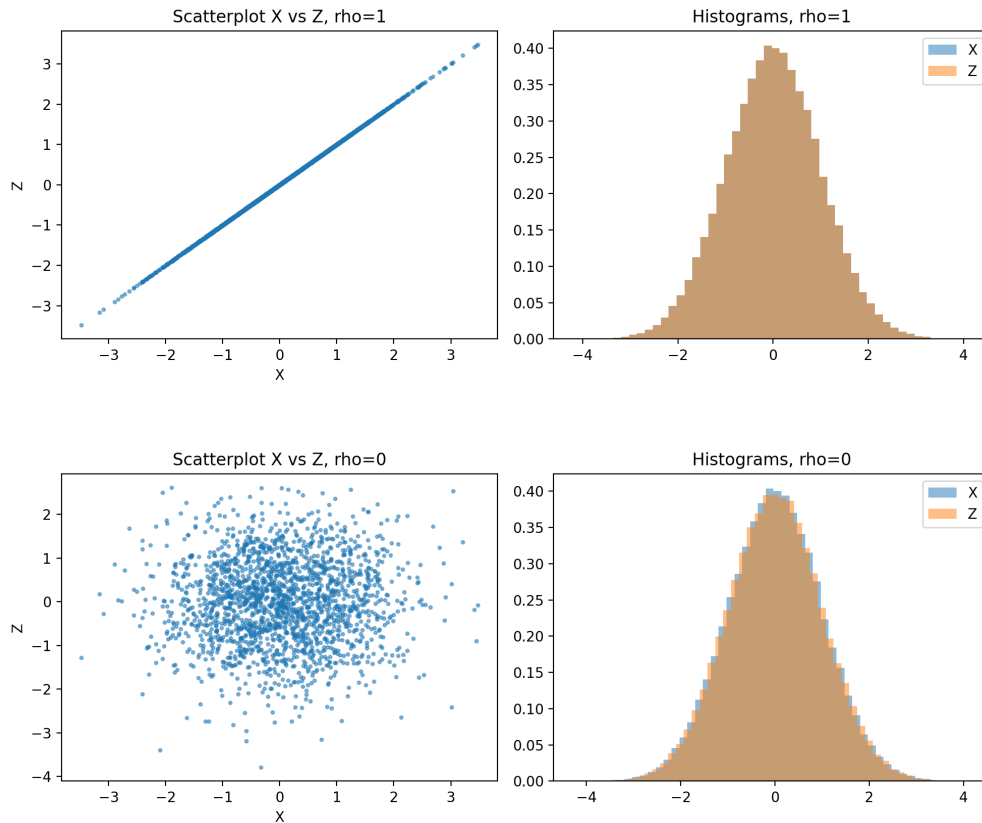
since X and Y are independent with $\text{Var}(X) = \text{Var}(Y) = 1$ and $\text{Cov}(X, Y) = 0$. Hence $Z_\rho \sim N(0, 1)$, so (X, Z_ρ) is a coupling of two standard Gaussian random variables for every $\rho \in [-1, 1]$.

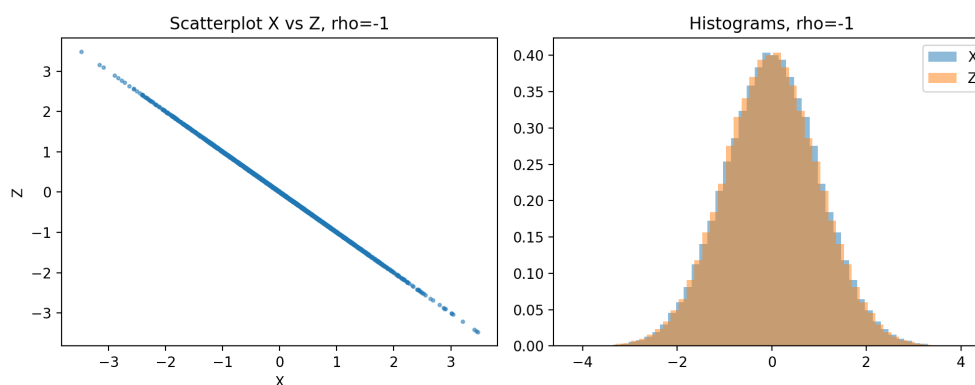
For a numerical verification, we simulate $n = 10^5$ i.i.d. pairs (X_i, Y_i) from $N(0, 1)$ and form

$$Z_{\rho,i} := \rho X_i + \sqrt{1 - \rho^2} Y_i, \quad i = 1, \dots, n,$$

for three values $\rho \in \{1, 0, -1\}$. The following R (or Python) code generates scatterplots of $(X_i, Z_{\rho,i})$ and overlaid histograms of the marginals X and Z_ρ .

The resulting figures are included below.





In each case, the histograms of X and Z_ρ are nearly indistinguishable and closely match the standard normal density, confirming that both marginals are approximately $N(0, 1)$. The scatterplots show sample correlation close to ρ , illustrating how this coupling interpolates between identical, independent, and antithetic Gaussian pairs.

3. Consider $(X_1, X_2) = (aX + bY, cX + dY)$. Find values of (a, b, c, d) such that:

- (a) (X_1, X_2) is a coupling of two standard Gaussian random variables;
- (b) $X_1 + X_2 = X$.

Remark: We may interpret X as a fixed Gaussian “budget” to be split between X_1 and X_2 , while requiring both parts to still be standard Gaussian.

Solution: Let

$$X_1 = aX + bY, \quad X_2 = cX + dY.$$

Since X and Y are independent standard normals, requiring X_1 and X_2 to be standard Gaussian gives

$$a^2 + b^2 = 1, \quad c^2 + d^2 = 1.$$

The condition $X_1 + X_2 = X$ implies

$$(a + c)X + (b + d)Y = X,$$

so

$$a + c = 1, \quad b + d = 0.$$

One convenient family of solutions is obtained as follows: choose any $b \in [-1, 1]$, set

$$a = \sqrt{1 - b^2}, \quad c = 1 - a, \quad d = -b.$$

Then

$$a^2 + b^2 = 1,$$

and also

$$a + c = a + (1 - a) = 1, \quad b + d = b - b = 0.$$

Finally,

$$c^2 + d^2 = (1 - a)^2 + b^2 = 1 - 2a + a^2 + b^2 = 1,$$

because $a^2 + b^2 = 1$. Hence this choice gives a valid coupling of two standard Gaussian random variables satisfying $X_1 + X_2 = X$.

Problem 2: Random walk on the complete graph

Consider the Markov chain on $\Omega = \{1, 2, \dots, n\}$ defined as follows: if the chain is currently at state x , then it moves uniformly to one of the other $n - 1$ states. In other words,

$$P(x, z) = \frac{1}{n-1} \mathbf{1}\{z \neq x\}.$$

1. Show that the uniform distribution on Ω is stationary for this Markov chain.

Solution: Let π be the uniform distribution on Ω , so $\pi(x) = 1/n$ for each x . For any $z \in \Omega$,

$$(\pi P)(z) = \sum_{x \in \Omega} \pi(x) P(x, z) = \sum_{x \neq z} \frac{1}{n} \cdot \frac{1}{n-1} = \frac{n-1}{n} \cdot \frac{1}{n-1} = \frac{1}{n} = \pi(z).$$

Therefore the uniform distribution is stationary.

2. Let $x \neq y$. Compute $\|P(x, \cdot) - P(y, \cdot)\|_{\text{TV}}$.

Solution: By definition,

$$\|P(x, \cdot) - P(y, \cdot)\|_{\text{TV}} = \frac{1}{2} \sum_{z \in \Omega} |P(x, z) - P(y, z)|.$$

The two distributions differ only at $z = x$ and $z = y$. Indeed,

$$|P(x, x) - P(y, x)| = \left| 0 - \frac{1}{n-1} \right| = \frac{1}{n-1},$$

and

$$|P(x, y) - P(y, y)| = \left| \frac{1}{n-1} - 0 \right| = \frac{1}{n-1}.$$

Thus

$$\|P(x, \cdot) - P(y, \cdot)\|_{\text{TV}} = \frac{1}{2} \left(\frac{1}{n-1} + \frac{1}{n-1} \right) = \frac{1}{n-1}.$$

3. Construct a coupling (X_1, Y_1) of $P(x, \cdot)$ and $P(y, \cdot)$ such that

$$\mathbb{P}(X_1 \neq Y_1) = \|P(x, \cdot) - P(y, \cdot)\|_{\text{TV}}.$$

Solution: Define a coupling of $P(x, \cdot)$ and $P(y, \cdot)$ as follows.

- With probability $\frac{1}{n-1}$, set $(X_1, Y_1) = (y, x)$.
- With the remaining probability $1 - \frac{1}{n-1}$, choose Z uniformly from $\Omega \setminus \{x, y\}$ and set $(X_1, Y_1) = (Z, Z)$.

We first check that the marginals are correct.

For the X -marginal:

$$\mathbb{P}(X_1 = y) = \frac{1}{n-1}, \quad \mathbb{P}(X_1 = x) = 0,$$

and for any $z \in \Omega \setminus \{x, y\}$,

$$\mathbb{P}(X_1 = z) = \left(1 - \frac{1}{n-1}\right) \mathbb{P}(Z = z) = \frac{n-2}{n-1} \cdot \frac{1}{n-2} = \frac{1}{n-1}.$$

Thus X_1 is uniform on $\Omega \setminus \{x\}$, so $X_1 \sim P(x, \cdot)$. A symmetric argument shows $Y_1 \sim P(y, \cdot)$.

Next, we compute the disagreement probability under this coupling. Disagreement occurs only in the first case, when $(X_1, Y_1) = (y, x)$. Hence

$$\mathbb{P}(X_1 \neq Y_1) = \frac{1}{n-1}.$$

From part (2), we already know that $\|P(x, \cdot) - P(y, \cdot)\|_{\text{TV}} = \frac{1}{n-1}$, so this coupling satisfies

$$\mathbb{P}(X_1 \neq Y_1) = \|P(x, \cdot) - P(y, \cdot)\|_{\text{TV}},$$

as required.

4. Now run this coupling at every step. Let

$$T := \inf\{t \geq 0 : X_t = Y_t\}.$$

Show that if $X_0 \neq Y_0$, then

$$\mathbb{P}(T > t) = \left(\frac{1}{n-1}\right)^t.$$

Solution: We run at every step the coupling from part (3), and define

$$T := \inf\{t \geq 0 : X_t = Y_t\}.$$

We show that for $x \neq y$,

$$\mathbb{P}(T > t) = \left(\frac{1}{n-1}\right)^t.$$

From the construction in part (3), whenever $X_t \neq Y_t$, the one-step coupling satisfies

$$\mathbb{P}(X_{t+1} \neq Y_{t+1} \mid X_t \neq Y_t) = \frac{1}{n-1}.$$

Moreover, once the chains meet, the coupling keeps them together forever: if $X_t = Y_t$ then $X_{t'} = Y_{t'}$ for all $t' \geq t$.

We now prove the formula by induction on t .

Base case. For $t = 0$ and $x \neq y$ we have $T > 0$ almost surely, so $\mathbb{P}(T > 0) = 1 = (1/(n-1))^0$.

Inductive step. Assume $\mathbb{P}(T > t) = (1/(n-1))^t$ for some $t \geq 0$. Then

$$\mathbb{P}(T > t+1) = \mathbb{P}(T > t+1 \mid T > t) \mathbb{P}(T > t) = \mathbb{P}(X_{t+1} \neq Y_{t+1} \mid X_t \neq Y_t) \mathbb{P}(T > t),$$

where we used that $T > t$ is equivalent to $X_t \neq Y_t$ and the Markov property of the coupled chain. By the one-step calculation above,

$$\mathbb{P}(X_{t+1} \neq Y_{t+1} \mid X_t \neq Y_t) = \frac{1}{n-1},$$

so

$$\mathbb{P}(T > t+1) = \frac{1}{n-1} \cdot \left(\frac{1}{n-1}\right)^t = \left(\frac{1}{n-1}\right)^{t+1}.$$

This completes the induction and proves the claim.

5. Use the coupling inequality to conclude that

$$\max_{x,y} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}} \leq \left(\frac{1}{n-1}\right)^t.$$

Solution: By the coupling inequality,

$$\|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}} \leq \mathbb{P}(T > t).$$

Using the coupling from the previous parts,

$$\mathbb{P}(T > t) = \left(\frac{1}{n-1}\right)^t.$$

Hence

$$\max_{x,y} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}} \leq \left(\frac{1}{n-1}\right)^t.$$

Problem 3: Random-to-top shuffle

Consider a deck of n cards labeled $1, \dots, n$. A shuffling is a permutation of $\{1, 2, \dots, n\}$. For example, if $n = 5$, then $(5, 1, 4, 3, 2)$ means the fifth card is on the top of the deck, followed by the first card, and so on. We say a shuffle mixes the deck if, after a number of steps, the distribution of the permutation is close to the uniform distribution over all $n!$ permutations.

The random-to-top shuffle is the following Markov chain on permutations: at each step, choose one card label uniformly at random from $\{1, \dots, n\}$, remove that card from its current position, and move it to the top of the deck.

1. Show that the uniform distribution on all $n!$ permutations is stationary for this Markov chain.

Solution: Let π be the uniform distribution on S_n , so $\pi(\sigma) = \frac{1}{n!}$ for each $\sigma \in S_n$. We verify the balance equation $\pi P = \pi$ directly.

Fix any $\tau \in S_n$. We compute

$$(\pi P)(\tau) = \sum_{\sigma \in S_n} \pi(\sigma) P(\sigma, \tau).$$

We need to identify the permutations σ for which $P(\sigma, \tau) > 0$, i.e., those from which a single random-to-top move can produce τ . Such a σ is obtained by taking τ and inserting its top card back into one of the n positions in the deck. There are exactly n such permutations $\sigma_1, \dots, \sigma_n$, and for each one the transition probability is $P(\sigma_k, \tau) = \frac{1}{n}$ (the card on top of τ is chosen with probability $\frac{1}{n}$).

Therefore,

$$(\pi P)(\tau) = \sum_{k=1}^n \pi(\sigma_k) \cdot P(\sigma_k, \tau) = \sum_{k=1}^n \frac{1}{n!} \cdot \frac{1}{n} = \frac{n}{n \cdot n!} = \frac{1}{n!} = \pi(\tau).$$

Hence $\pi P = \pi$, so the uniform distribution is stationary.

2. Let X_t and Y_t be two copies of the random-to-top shuffle, started from two possibly different initial decks X_0 and Y_0 . Couple them by choosing the same card label at every step and moving that label to the top in both decks.

Let A_t be the set of card labels that have been chosen at least once by time t . Show that the relative order of the cards in A_t is the same in X_t and Y_t .

Solution: We prove the claim by induction on t . For $t = 0$, the set A_0 is empty, so the statement is trivial.

Assume the claim holds at time t , and consider time $t + 1$. Let i_{t+1} be the label chosen at step $t + 1$.

Case 1: $i_{t+1} \in A_t$. Then in both decks we move the same card i_{t+1} to the top. In each deck, the positions of all other labels in A_t do not change; only i_{t+1} moves above them. Since the relative order of the labels in A_t was the same in X_t and Y_t by the induction hypothesis, and we apply the same operation to i_{t+1} in both decks, the relative order of the labels in A_t remains the same in X_{t+1} and Y_{t+1} . In this case $A_{t+1} = A_t$.

Case 2: $i_{t+1} \notin A_t$. Then $A_{t+1} = A_t \cup \{i_{t+1}\}$, and in both decks the new label i_{t+1} is moved to the top of the deck. In particular, i_{t+1} is placed above all labels in A_t in both X_{t+1} and Y_{t+1} , while the relative order of the labels in A_t among themselves is unchanged. Thus the relative order of the labels in A_{t+1} is the same in X_{t+1} and Y_{t+1} .

In either case, the relative order of the labels in A_{t+1} is the same in the two decks. By induction, the claim holds for all t .

3. Show that if $A_t = \{1, \dots, n\}$, then $X_t = Y_t$. Let

$$T := \inf\{t \geq 0 : X_t = Y_t\}.$$

Deduce that

$$\mathbb{P}(T > t) \leq \mathbb{P}(A_t \neq \{1, \dots, n\}).$$

Solution: If $A_t = \{1, \dots, n\}$, then every label has been selected at least once. By the previous part, the relative order of all labels is then the same in X_t and Y_t , so $X_t = Y_t$.

Therefore, if $T > t$, then necessarily $A_t \neq \{1, \dots, n\}$. Hence

$$\mathbb{P}(T > t) \leq \mathbb{P}(A_t \neq \{1, \dots, n\}).$$

4. Use the union bound to prove that

$$\mathbb{P}(A_t \neq \{1, \dots, n\}) \leq n \left(1 - \frac{1}{n}\right)^t \leq ne^{-t/n}.$$

Solution: For each label i , let E_i be the event that label i has not been chosen by time t . Then

$$\mathbb{P}(E_i) = \left(1 - \frac{1}{n}\right)^t.$$

If $A_t \neq \{1, \dots, n\}$, then at least one E_i occurs. By the union bound,

$$\mathbb{P}(A_t \neq \{1, \dots, n\}) \leq \sum_{i=1}^n \mathbb{P}(E_i) = n \left(1 - \frac{1}{n}\right)^t.$$

Using $1 - 1/n \leq e^{-1/n}$ gives

$$\mathbb{P}(A_t \neq \{1, \dots, n\}) \leq ne^{-t/n}.$$

5. Using the coupling inequality, conclude that

$$\max_{x,y} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}} \leq ne^{-t/n}.$$

In particular, show that the mixing time is at most

$$t_{\text{mix}}(\varepsilon) \leq n \log n + n \log(1/\varepsilon).$$

Solution: By the coupling inequality and the previous bounds,

$$\max_{x,y} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}} \leq \mathbb{P}(T > t) \leq \mathbb{P}(A_t \neq \{1, \dots, n\}) \leq ne^{-t/n}.$$

To make this at most ε , it suffices that

$$ne^{-t/n} \leq \varepsilon.$$

Taking logarithms yields

$$t \geq n \log n + n \log(1/\varepsilon).$$

Therefore,

$$t_{\text{mix}}(\varepsilon) \leq n \log n + n \log(1/\varepsilon).$$

Problem 4: Conditional Bernoulli distribution

In this problem, we study an MCMC algorithm for sampling from the conditional Bernoulli distribution. Fix two positive integers $0 < I < N$ and parameters $(p_1, \dots, p_N) \in (0, 1)^N$. The conditional Bernoulli distribution is the distribution on binary vectors $\{0, 1\}^N$ defined by

$$\mathbb{P}(X_1 = x_1, \dots, X_N = x_N) \propto \prod_{k=1}^N p_k^{x_k} (1 - p_k)^{1-x_k} \mathbf{1}\left(\sum_{i=1}^N x_i = I\right).$$

Equivalently, it is the distribution of independent Bernoulli random variables $X_k \sim \text{Ber}(p_k)$ conditioned on $\sum_{i=1}^N X_i = I$.

We now consider the following MCMC algorithm. Given a current state $x^{(t)} \in \{0, 1\}^N$ satisfying $\sum_{i=1}^N x_i^{(t)} = I$, define

$$S_0(t) := \{i : x_i^{(t)} = 0\}, \quad S_1(t) := \{j : x_j^{(t)} = 1\}.$$

At each iteration, choose $i_t \in S_0(t)$ and $j_t \in S_1(t)$ independently and uniformly at random. The proposal flips the two coordinates $x_{i_t}^{(t)}$ and $x_{j_t}^{(t)}$, and is then accepted or rejected according to the Metropolis–Hastings rule.

1. Calculate the acceptance probability for a proposed flip of coordinates i_t and j_t .

Solution: Let $x \in \mathcal{X}$ be the current state and let y be obtained by changing $x_i = 0$ to $y_i = 1$ and $x_j = 1$ to $y_j = 0$, for some $i \in S_0(x)$ and $j \in S_1(x)$. Since $|S_0(x)| = N - I$ and $|S_1(x)| = I$, the proposal probability is

$$q(x, y) = \frac{1}{|S_0(x)|} \frac{1}{|S_1(x)|} = \frac{1}{(N - I) I}.$$

The reverse move $y \rightarrow x$ has the same proposal probability, so $q(y, x) = q(x, y)$. Therefore the Metropolis–Hastings acceptance probability is

$$\alpha(x, y) = \min\left\{1, \frac{\pi(y)}{\pi(x)}\right\}.$$

Since π is proportional to $\prod_{k=1}^N p_k^{x_k} (1-p_k)^{1-x_k}$ restricted to $\sum_{k=1}^N x_k = I$, the normalizing constant cancels in the ratio, and only coordinates i and j differ between x and y . Thus

$$\frac{\pi(y)}{\pi(x)} = \frac{p_i^{y_i} (1-p_i)^{1-y_i} p_j^{y_j} (1-p_j)^{1-y_j}}{p_i^{x_i} (1-p_i)^{1-x_i} p_j^{x_j} (1-p_j)^{1-x_j}} = \frac{p_i}{1-p_i} \cdot \frac{1-p_j}{p_j}.$$

Hence

$$\alpha(x, y) = \min \left\{ 1, \frac{p_i}{1-p_i} \cdot \frac{1-p_j}{p_j} \right\}.$$

2. Prove that the Markov chain is irreducible on

$$\mathcal{X} := \left\{ x \in \{0, 1\}^N : \sum_{i=1}^N x_i = I \right\}.$$

Solution: Take any $x, y \in \mathcal{X}$. Let

$$A = \{k : x_k = 1, y_k = 0\}, \quad B = \{k : x_k = 0, y_k = 1\}.$$

Since both x and y have exactly I ones, we have $|A| = |B|$. Pair the elements of A and B arbitrarily, say $(a_1, b_1), \dots, (a_m, b_m)$ with $m = |A|$.

Starting from x , for $\ell = 1, \dots, m$ perform the following move: set the coordinate at a_ℓ from 1 to 0 and the coordinate at b_ℓ from 0 to 1. Each such move is exactly of the allowed form in the Markov chain (flipping one 0 to 1 and one 1 to 0), so every intermediate state lies in $\mathcal{X}$ and has exactly I ones.

After performing all m moves, we obtain y , because all coordinates where x and y differ have been corrected. Each of these moves has strictly positive proposal probability and strictly positive acceptance probability since every $p_k \in (0, 1)$. Hence there is a positive-probability path from x to y , so the chain is irreducible on $\mathcal{X}$.

3. Assume there exist constants $0 < c < C < \infty$ such that

$$c < \frac{p_k}{1-p_k} < C \quad \text{for every } k = 1, \dots, N.$$

Prove that the above algorithm has mixing time

$$t_{\text{mix}}(\varepsilon) = O(N \log N).$$

Solution: We use path coupling. Define a graph on $\mathcal{X}$ where $x \sim y$ if they differ in exactly two coordinates (one $0 \rightarrow 1$ and one $1 \rightarrow 0$), and let

$$\ell(x, y) = \frac{1}{2} \sum_{k=1}^N |x_k - y_k|$$

be the (Hamming) distance. It suffices to consider neighboring states x, y with $\ell(x, y) = 1$, i.e., there exist unique indices u, v such that

$$x_u = 1, \quad x_v = 0, \quad y_u = 0, \quad y_v = 1,$$

and $x_k = y_k$ for all $k \neq u, v$.

Coupled proposal. At each step, draw the same pair (i_t, j_t) , with $i_t \in S_0(x)$, $j_t \in S_1(x)$ for the x -chain, and couple the y -chain so that it uses the same pair whenever valid (i.e. when $i_t \in S_0(y)$ and $j_t \in S_1(y)$ as well). Since x and y differ only at u, v , we have

$$S_0(x) \triangle S_0(y) = \{u, v\}, \quad S_1(x) \triangle S_1(y) = \{u, v\}.$$

Drift computation. There are three types of proposed moves:

- *Coalescing move* $(i_t, j_t) = (v, u)$: this is the unique proposal that swaps the disagreement and makes the two chains coincide. It is chosen with probability

$$\frac{1}{|S_0(x)| |S_1(x)|} = \frac{1}{(N - I) I}.$$

The Metropolis–Hastings acceptance probability for this move in the x -chain is

$$\alpha(x, x') = \min \left\{ 1, \frac{p_v}{1 - p_v} \cdot \frac{1 - p_u}{p_u} \right\}.$$

By the assumption $c < p_k / (1 - p_k) < C$ for all k , the ratio inside the minimum lies in $(c/C, C/c)$, so there exists $\alpha_0 := \min(1, c/C) > 0$ such that $\alpha(x, x') \geq \alpha_0$. If both chains accept this move, ℓ decreases by 1. Thus this case contributes at most

$$-\frac{\alpha_0}{(N - I) I}$$

to the expected change in ℓ .

- *Neutral moves* (neither i_t nor j_t is in $\{u, v\}$): the same coordinates are proposed and updated in both chains, and the accept/reject decisions are the same under our coupling. Hence $\ell(X_1, Y_1) = \ell(X_0, Y_0)$, and these moves contribute 0 to the drift.

- *Discordant moves* (exactly one of i_t, j_t lies in $\{u, v\}$): in this case the two chains may update different coordinates. There are $O(N)$ such pairs (i_t, j_t) , each chosen with probability at most $1/((N - I)I)$, and in the worst case ℓ can increase by at most 1. Thus the total positive contribution to the drift from discordant moves is bounded above by

$$\frac{C_1}{(N - I)I}$$

for some constant C_1 independent of N .

Putting these together, for neighboring x, y we obtain

$$\mathbb{E}[\ell(X_1, Y_1) - \ell(X_0, Y_0) \mid X_0 = x, Y_0 = y] \leq -\frac{\alpha_0 - C_1}{(N - I)I}.$$

Since $I(N - I) \asymp N$ and α_0 depends only on c, C , by absorbing constants we can choose $\kappa > 0$ (depending only on c, C) such that, for all sufficiently large N ,

$$\mathbb{E}[\ell(X_1, Y_1) \mid X_0 = x, Y_0 = y] \leq \ell(x, y) \left(1 - \frac{\kappa}{N}\right) = 1 - \frac{\kappa}{N}.$$

Conclusion. By the path coupling theorem, for any $x, y \in \mathcal{X}$ and any $t \geq 0$,

$$\mathbb{E}[\ell(X_t, Y_t)] \leq \ell(x, y) \left(1 - \frac{\kappa}{N}\right)^t \leq N \left(1 - \frac{\kappa}{N}\right)^t \leq N e^{-\kappa t/N}.$$

Since

$$\|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}} \leq \mathbb{P}(X_t \neq Y_t) \leq \mathbb{E}[\ell(X_t, Y_t)],$$

we obtain

$$\max_{x, y} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}} \leq N e^{-\kappa t/N}.$$

Solving $N e^{-\kappa t/N} \leq \varepsilon$ for t gives

$$t \geq \frac{N}{\kappa} (\log N + \log(1/\varepsilon)),$$

so the mixing time satisfies

$$t_{\text{mix}}(\varepsilon) = O(N \log N).$$